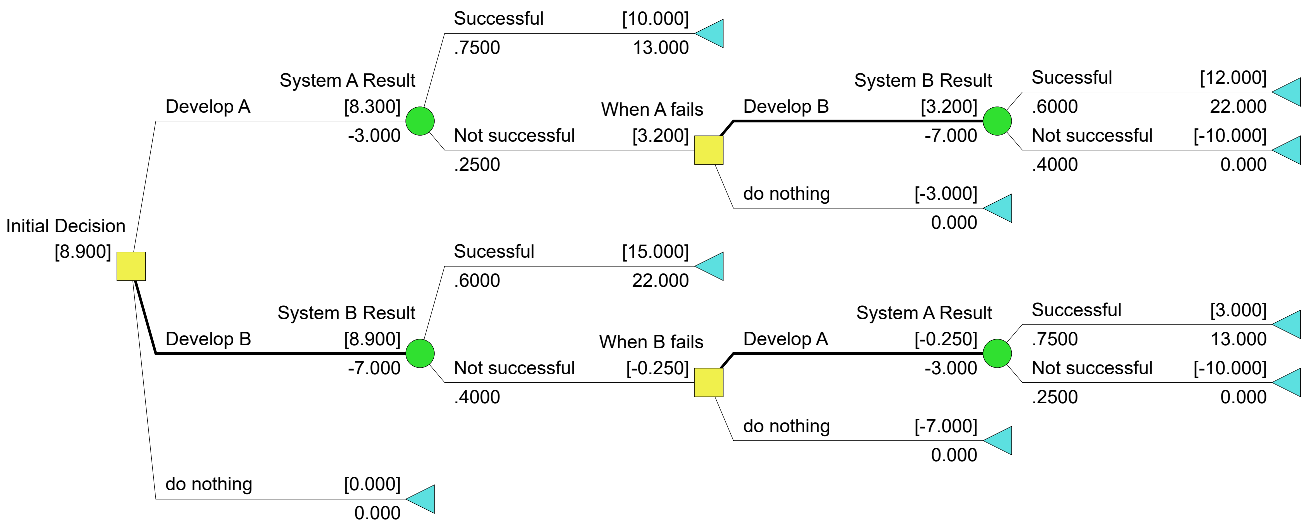
IE5203 Decision Analysis
Solutions to Assignment 1

Question 1.

Decision Tree:



Optimal Decision Policy:

Develop System B
If successful
    Manufacture and market System B
Else
    Develop System A
    If successful
        Manufacture and market System A
    Else
        Do nothing.

- Certainty equivalent = Expected Profit = **$ <u>8.900 million</u>**

Question 2.

Risk Profiles in PMF for the three initial alternatives:

Develop A

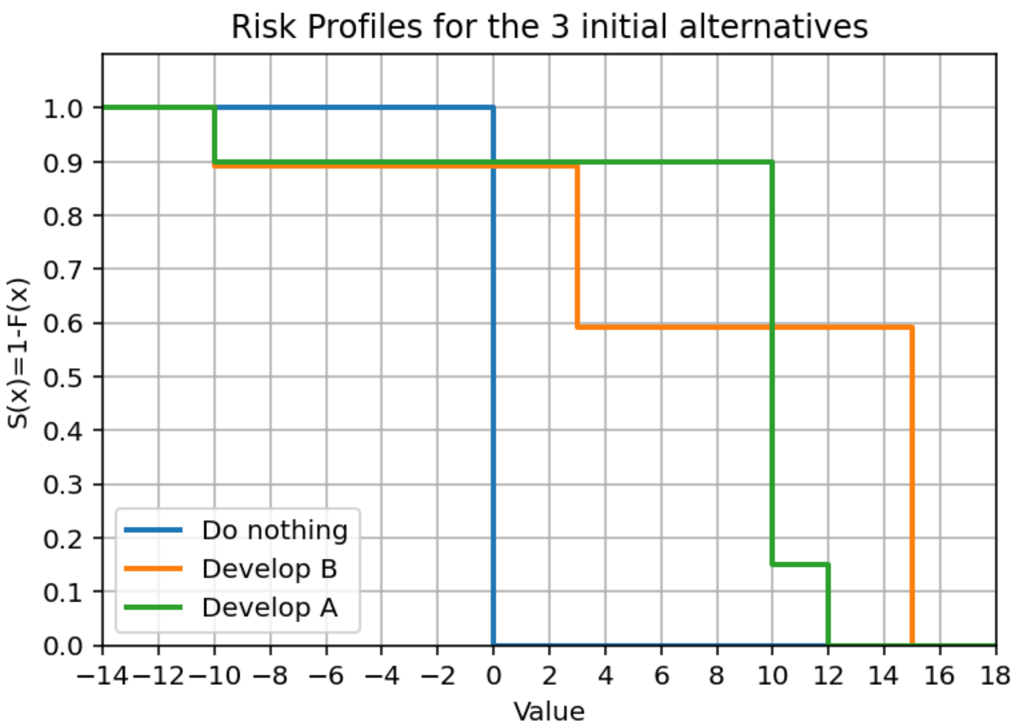
	End-point value	Probability
1	-10	$0.25 \times 0.4 = 0.1$
2	10	0.75
3	12	$0.25 \times 0.6 = 0.15$

Develop B

	End-point value	Probability
1	-10	$0.4 \times 0.25 = 0.1$
2	3	$0.4 \times 0.75 = 0.3$
3	15	0.6

Do nothing

	End-point value	Probability
1	0	1



Question 3.

There is no first-order stochastic dominance, as all the 3 excess probability distributions overlap.

Do nothing does not second-order stochastically dominate Develop A

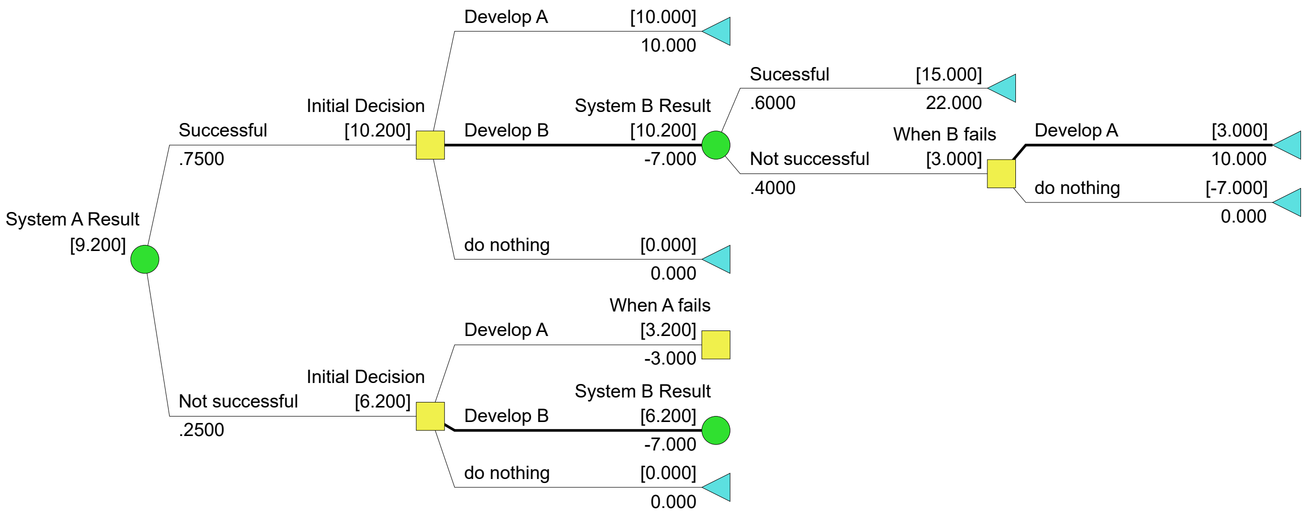
Do nothing does not second-order stochastically dominate Develop B

Develop A does not second-order stochastically dominate Develop B

Hence, there is no second-order stochastic dominance among all the 3 alternatives.

Question 4.

Decision model with free perfect information on Design A result:



- CE with free perfect information on the R&D result of System A = $9.200 million
- CE with no information = $8.900 million (from base model)

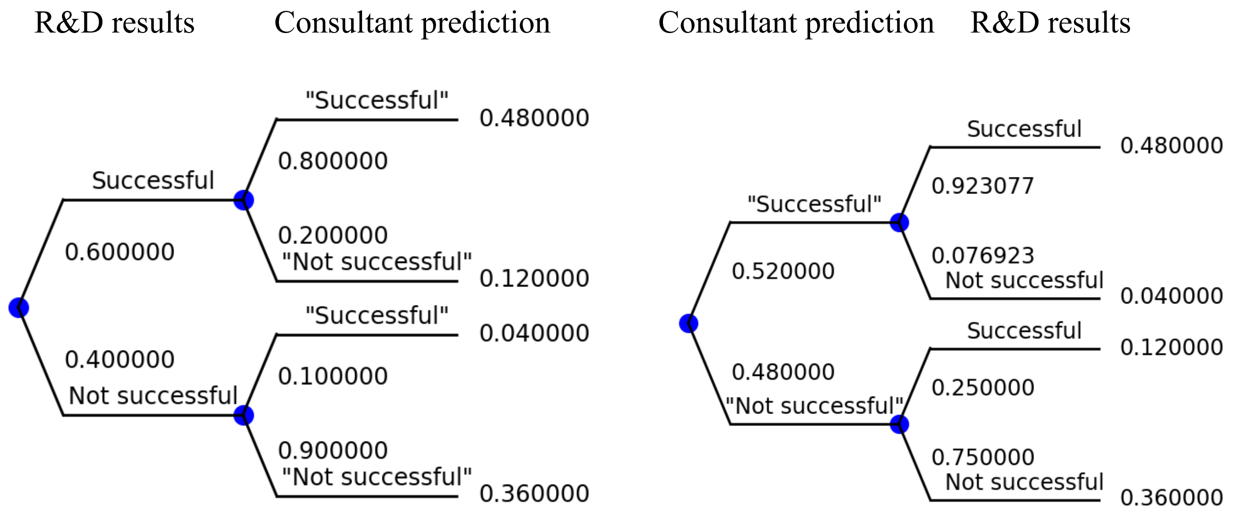
- Hence, the expected value of perfect information on the R&D result of System A

  = $9.200 - $8,900

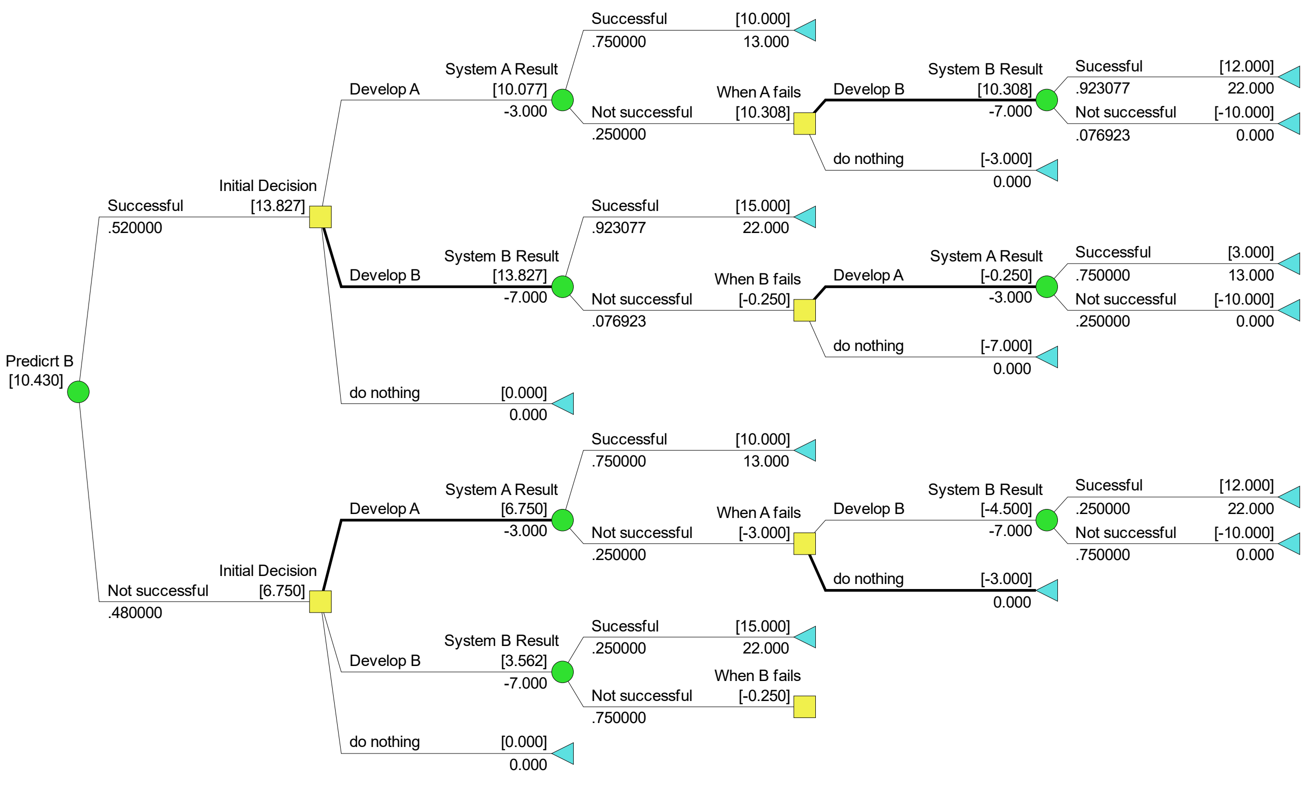
  = **<u>$ 0.300 million</u>**

Question 5

Flipping the tree on the consultant's prediction of the R&D result of System B:



Decision Model with free prediction of the R&D result of System B:



- CE with free prediction of the R&D result of System B = $10.430 million
- CE with no prediction = $8.900 million
- Expected value of imperfection information = 10,430 – 8.900 = $1.530 million

(*a*) Optimal Decision Policy

```
If X > $1.530 million
    Do not engage the consultant.
    Develop System B
    If successful
        Manufacture and market System B
    Else // System B not successful
        Develop System A
        If successful
            Manufacture and market System A
        Else
             Do nothing (exit)

Else    // X ≤ $1.530 million
    Engage the consultant for $X
    If the prediction is that System B is successful
        Develop System B
        If successful
            Produce and market System B
        Else  // System B is not successful
            Develop System A
            If successful
                Produce and market System A
            Else
                Do nothing (exit)
    Else  // prediction is that System B is unsuccessful
        Develop System A
        If successful
            Produce and market System A
        Else    // System A not successful
            Do nothing (exit)
```

(*b*) Certainty Equivalent of Optimal Policy

CE of optimal policy

= max(CE with no prediction, CE with free prediction – X)

= max($8.900 million, $10.430 million – X)